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1  def c_scan(requests, head, disk_size): 1 usage
2      left = []
3      right = []
4
5      for req in requests:
6          if req < head:
7              left.append(req)
8          else:
9              right.append(req)
10
11     left.sort()
12     right.sort()
13
14     seek_sequence = []
15     total_seek = 0
16     current = head
17
18     # Service right side requests
19     for track in right:
20         total_seek += abs(current - track)
21         current = track
22         seek_sequence.append(track)
23
24     # Move to end of disk
25     total_seek += abs(current - (disk_size - 1))
26     current = disk_size - 1
27
28     # Jump to beginning

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27
28     # Jump to beginning
29     total_seek += (disk_size - 1)
30     current = 0
31
32     # Service left side requests
33     for track in left:
34         total_seek += abs(current - track)
35         current = track
36         seek_sequence.append(track)
37
38     print("\nSeek Sequence:")
39     print(" -> ".join(map(str, seek_sequence)))
40
41     print("Total Head Movement =", total_seek)
42
43
44     # Input
45     requests = list(map(int, input("Enter disk requests: ").split()))
46     head = int(input("Enter initial head position: "))
47     disk_size = int(input("Enter disk size: "))
48
49     c_scan(requests, head, disk_size)

```

Enter disk requests: 20 23 23 45 67 89

Enter initial head position: 55

Enter disk size: 3

Seek Sequence:

67 -> 89 -> 20 -> 23 -> 23 -> 45

Total Head Movement = 168

Process finished with exit code 0